

TA 211

Manufacturing of Infinity Gauntlet

Instructor

Prof. Shashank Shekhar

Lab In-Charge

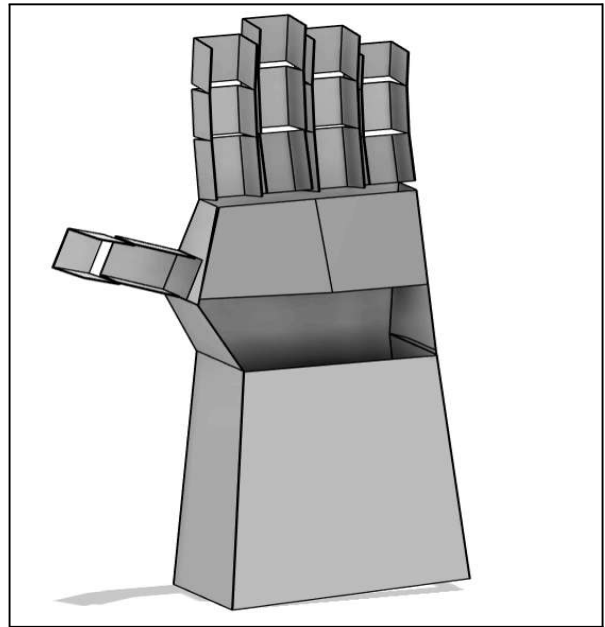
Mr Rakesh Dixit

Tutors

Adam Musa

Suraj Kumar

Raghav Mundra



Members' Names (Group M7 Monday)

1. 240444 Harshwardhan Giri Goswami
2. 240397 Gautam Prajapat
3. 240473 Ishanvi Salar
4. 220078 Aditya Raj Mishra
5. 240514 Kandolu Soni Sowbhagya
6. 240417 Gunjan

Content

S No.	Content	Page no.
1.	Certificate of Plagiarism	
2.	Introduction	
3.	Motivation	
4.	Acknowledgement	
5.	Work Distribution	
6.	Material List	
7.	Process used in making parts	
8.	Isometric View	
9.	Part Drawings	
10.	Final Model	

Certificate of Plagiarism

We hereby declare that the project titled Manufacturing of Infinity Gauntlet is an original work carried out by us. The content presented in this report has not been copied from any other source without proper citation and acknowledgement. All references used have been duly mentioned, and the work complies with the institution's guidelines on academic honesty and plagiarism.

We understand that plagiarism is a serious academic offence and affirm that this submission is free from any such violations.

Introduction

This report details the initial steps and project plan for the development of an Infinity Gauntlet model, which will be designed and manufactured over the coming weeks as part of our Manufacturing Process (TA211) lab project.

The primary objective of this project is to apply the theoretical knowledge gained in the classroom to a practical, hands-on engineering task, allowing us to explore the intricacies of design, prototyping, and manufacturing processes. The Infinity Gauntlet model will be a scaled-down, functional representation inspired by the iconic glove concept. It will incorporate design features such as mobility for the fingers, embedded components, and an ergonomic structure to replicate both aesthetics and functionality.

In this project report, we have finalised the design, material selection, cost estimation, and work distribution. This report serves as the foundation for our project, outlining the goals, design strategies, and expected outcomes. It will guide our work in the coming weeks, ensuring a structured approach to the successful completion of the Infinity Gauntlet model.

Motivation

Our motivation for selecting the Infinity Gauntlet as our project stems from its iconic status as a symbol of power, creativity, and imagination in popular culture. The glove represents not only a fascinating fictional artefact but also an opportunity to merge aesthetics with engineering principles in its design and construction. By attempting to replicate the Infinity Gauntlet, even as a scaled and simplified model, we gain valuable insights into product design, ergonomics, and the integration of multiple components into a single structure.

This project challenges us to translate a conceptual idea into a tangible object, balancing both functionality and visual appeal. Furthermore, the project encourages teamwork, innovation, and hands-on problem-solving, while allowing us to apply the knowledge and skills acquired in this course. It serves as a creative yet practical exercise in manufacturing processes, enabling us to explore how engineering can bring even the most imaginative concepts to life.

Acknowledgement

We are deeply grateful to our tutor and course in-charge, **Prof. Shashank Shekhar** and **Lab In-Charge Mr. Rakesh Dixit**, for their invaluable guidance and support throughout the planning and execution of this project. His encouragement and insights have been instrumental in shaping our approach and deepening our understanding of the subject matter.

We would also like to express our sincere appreciation to all those who have supported us during this endeavour. Their contributions, whether through technical advice, thoughtful feedback, or moral support, have helped us navigate challenges and stay motivated.

Finally, we acknowledge the collaboration and effort of our team members, whose curiosity, persistence, and shared enthusiasm laid a strong foundation for this work. From troubleshooting shaders to analysing solid mechanics, every step has been a testament to our collective dedication and passion for learning.

Work Distribution

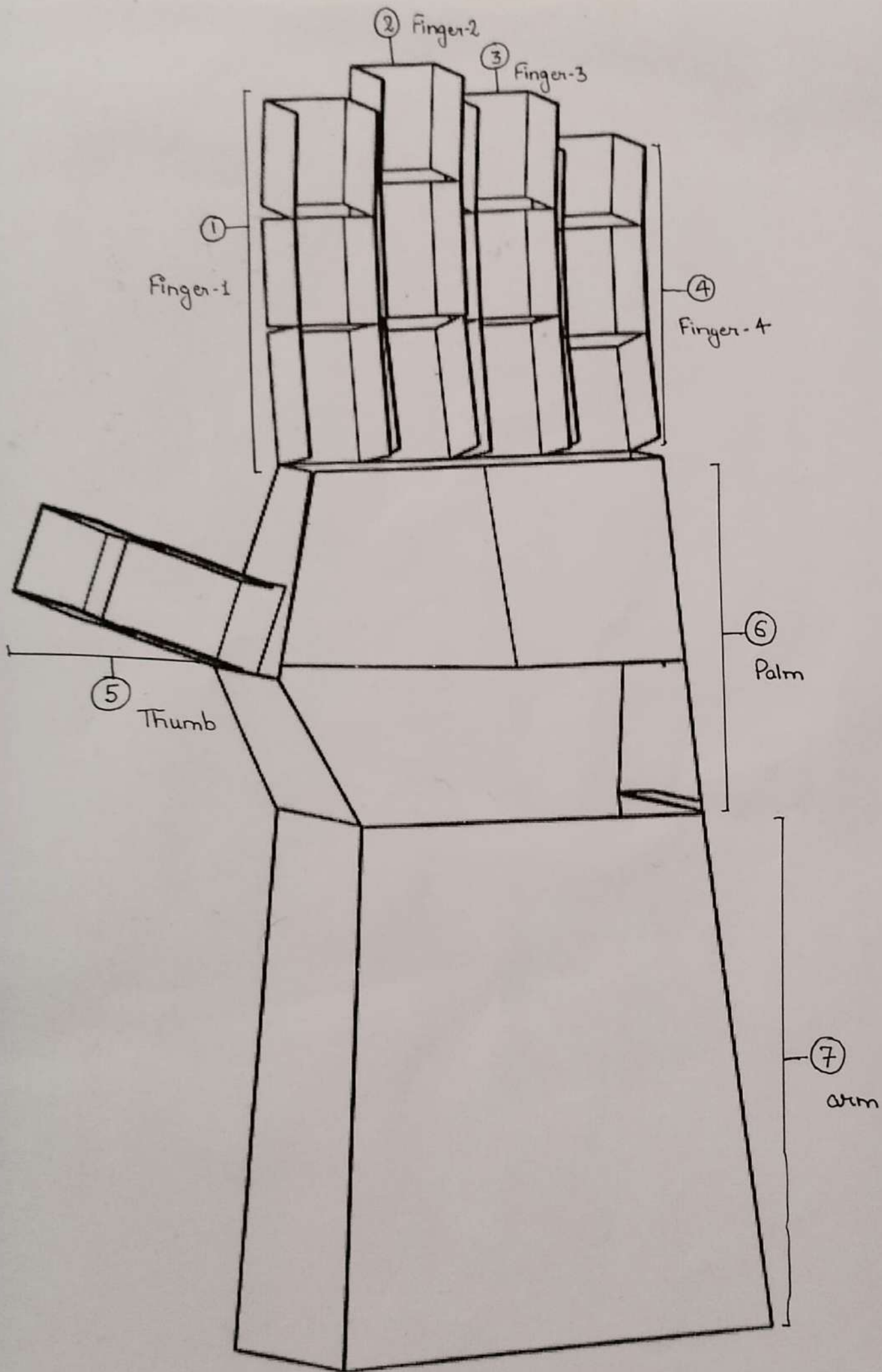
Name	Week 1	Week 2	Week3	Week 4	Week 5	Week 6
Gautam	fingers	fingers	Finger connection	Finger connection	Assembly	Assembly
Gunjan	fingers	fingers	Finger connection	Finger connection	Assembly	Assembly
Harshwardhan	palm	palm	Finger connection	Finger connection	Assembly	Assembly
Ishanvi	palm	palm	Palm connection	Palm connection	Assembly	Assembly
Soni	arm	arm	Palm connection	Palm connection	Assembly	Assembly
Aaditya	arm	arm	arm connection	arm connection	Assembly	Assembly

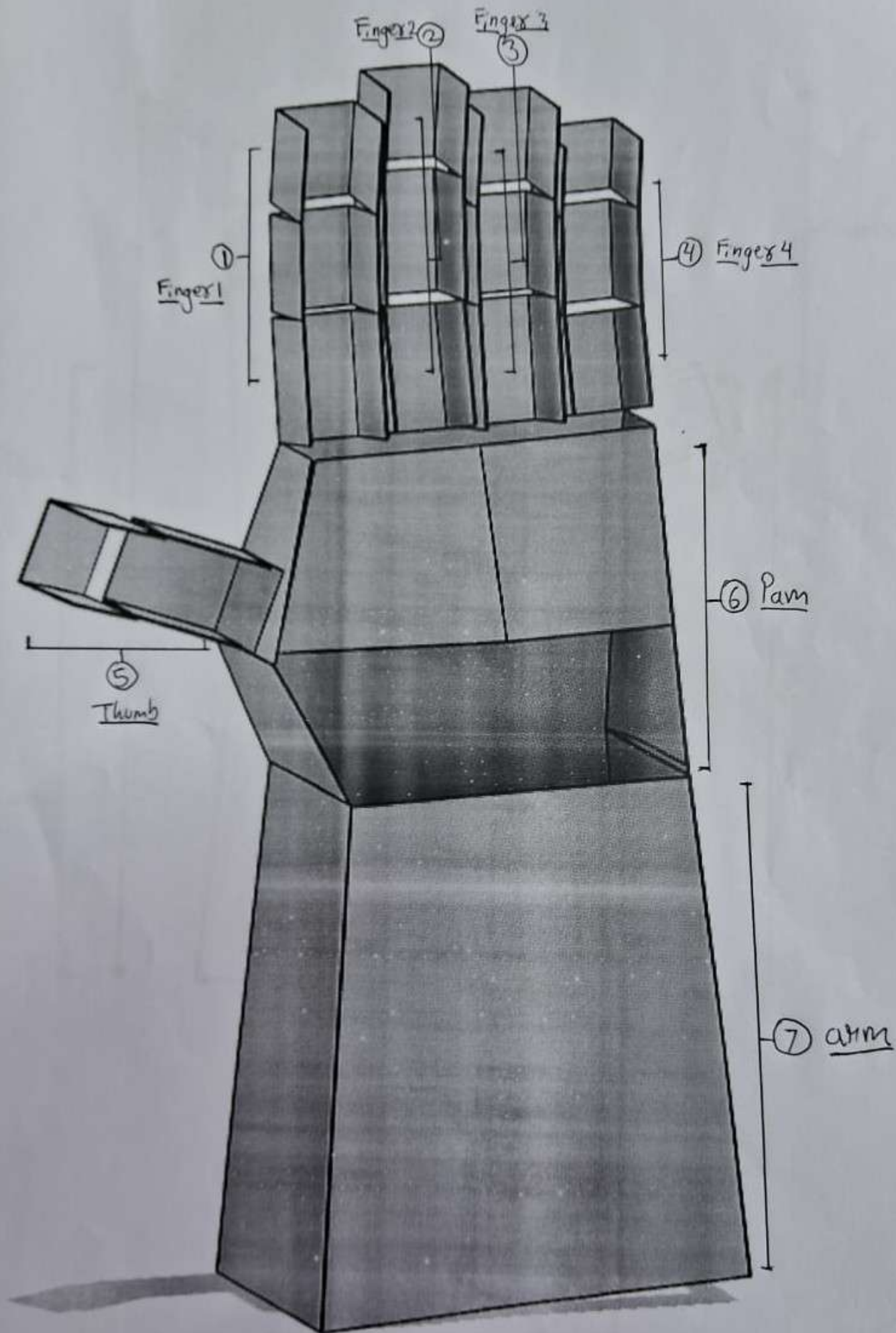
Material List

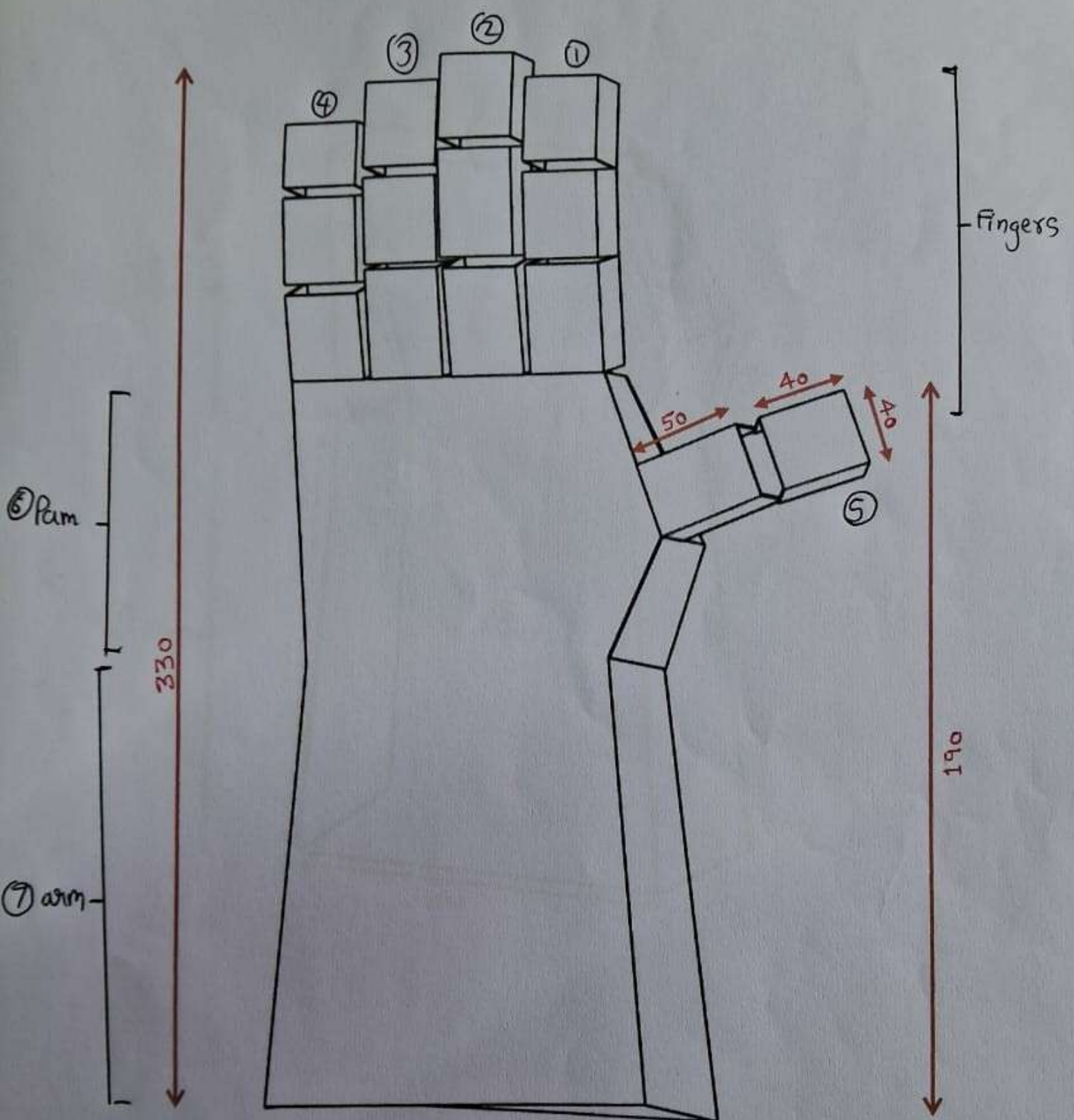
S.No	Part name	Material	Quantity	Required dimensions
1	Palm + Arm	Mild Steel Sheet (0.5mm)	1	200mm x 180mm base, 150mm top
2	Fingers	Mild Steel Sheet (1mm)	1	80mm x 30mm x 0.5mm
3	Rods	Mild Steel Round Rod(3mm Ø)	6	180mm length each
4	Joints	Nuts & Bolts	6	For connecting fingers and joints
5	Washers	Mild Steel Washers	15 pieces	Compatible with 3mm rod
6	Support Frame	Mild Steel Square Pipe	4 pieces	25mm x 25mm
7	Disks	Mild Steel Disks	3 pieces	3 each of 20mm, 30mm, and 40mm diameter

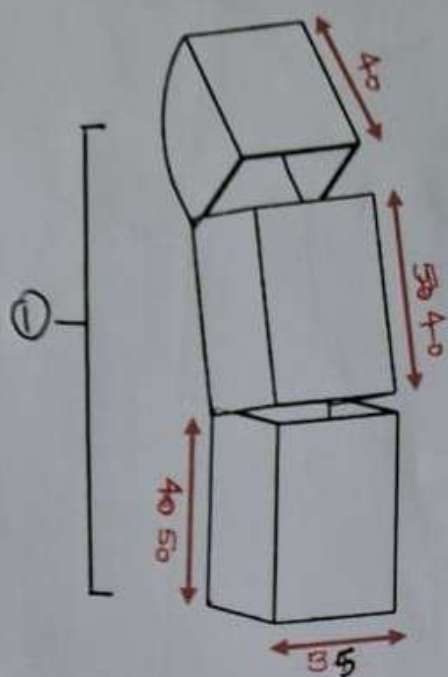
Process used in making parts

S No.	Parts	Process
1.	Finger 1	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
2.	Finger 2	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
3.	Finger 3	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
4.	Finger 4	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
5.	Thumb	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
6.	Pam	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication
7.	Arm	Sheet Metal Forming, Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication

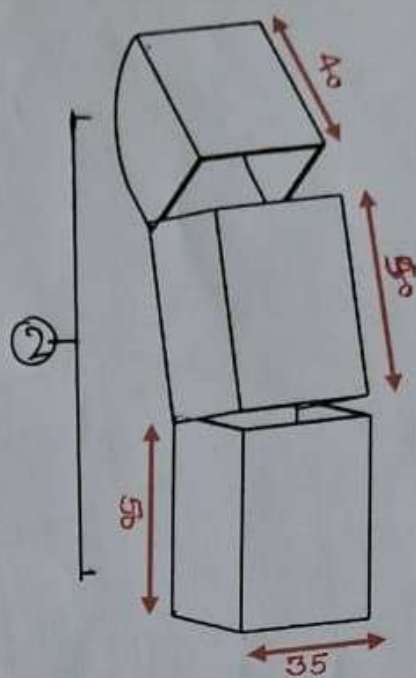




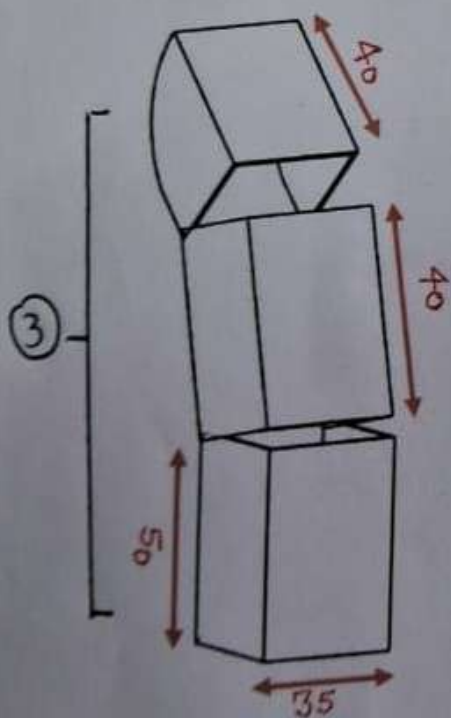




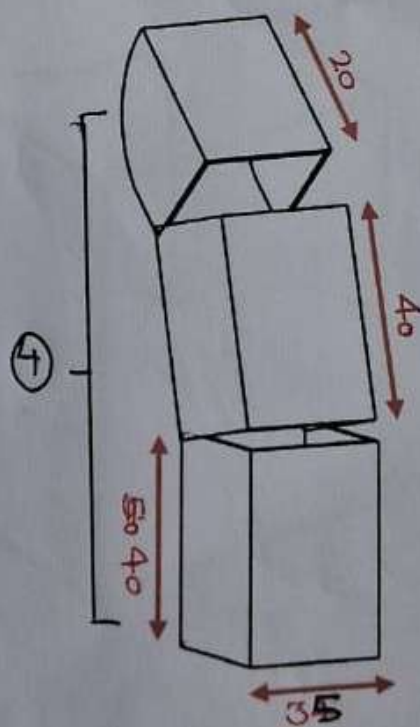
Index Finger 1



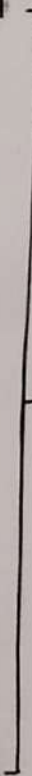
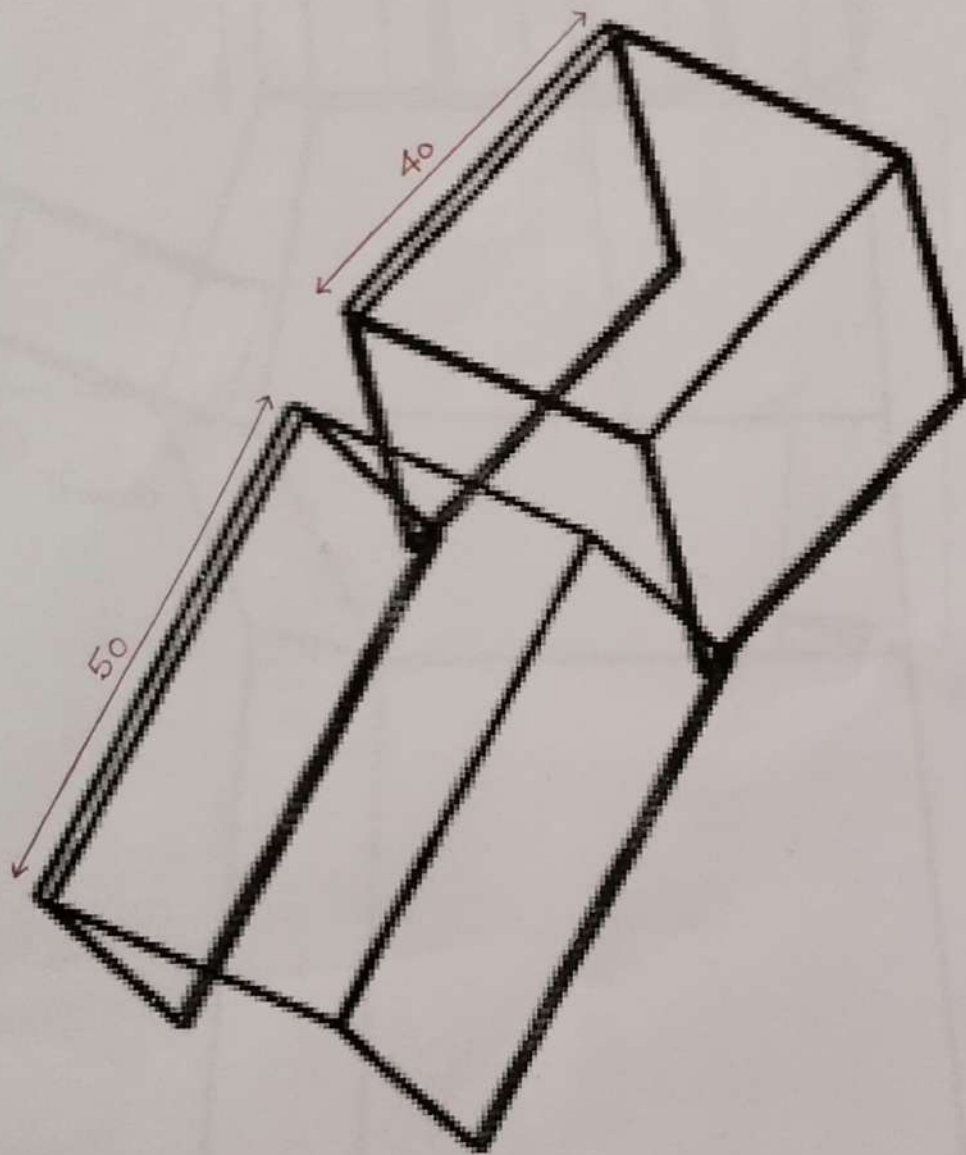
Middle Finger 2



Ring Finger 3



Little Finger 4

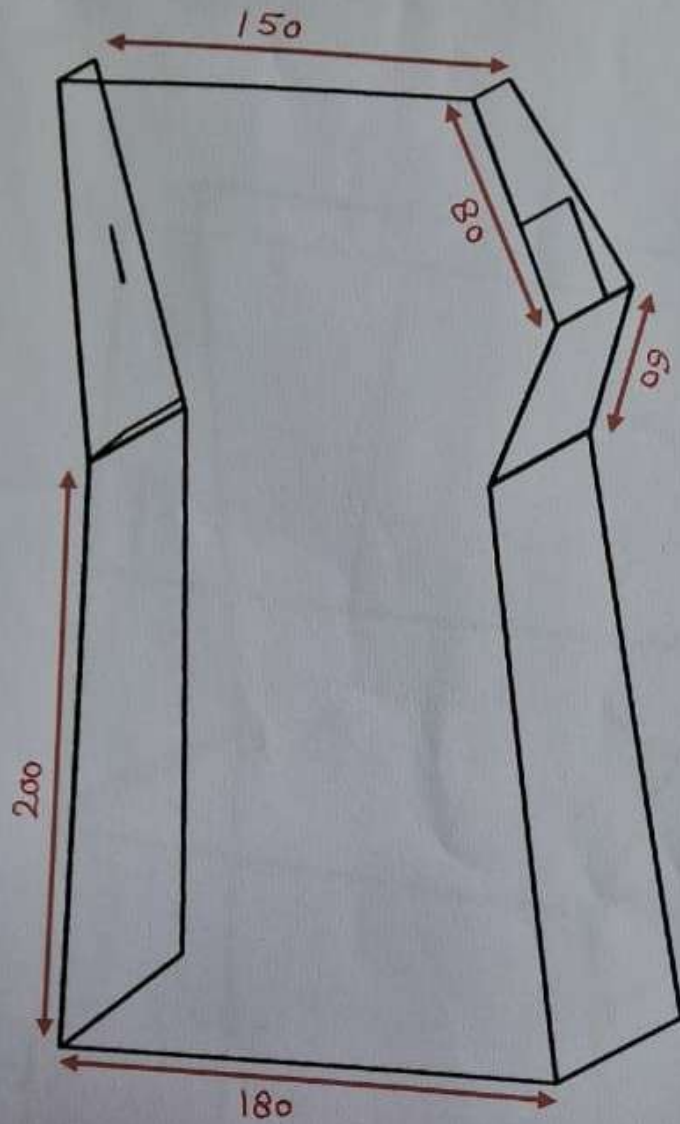


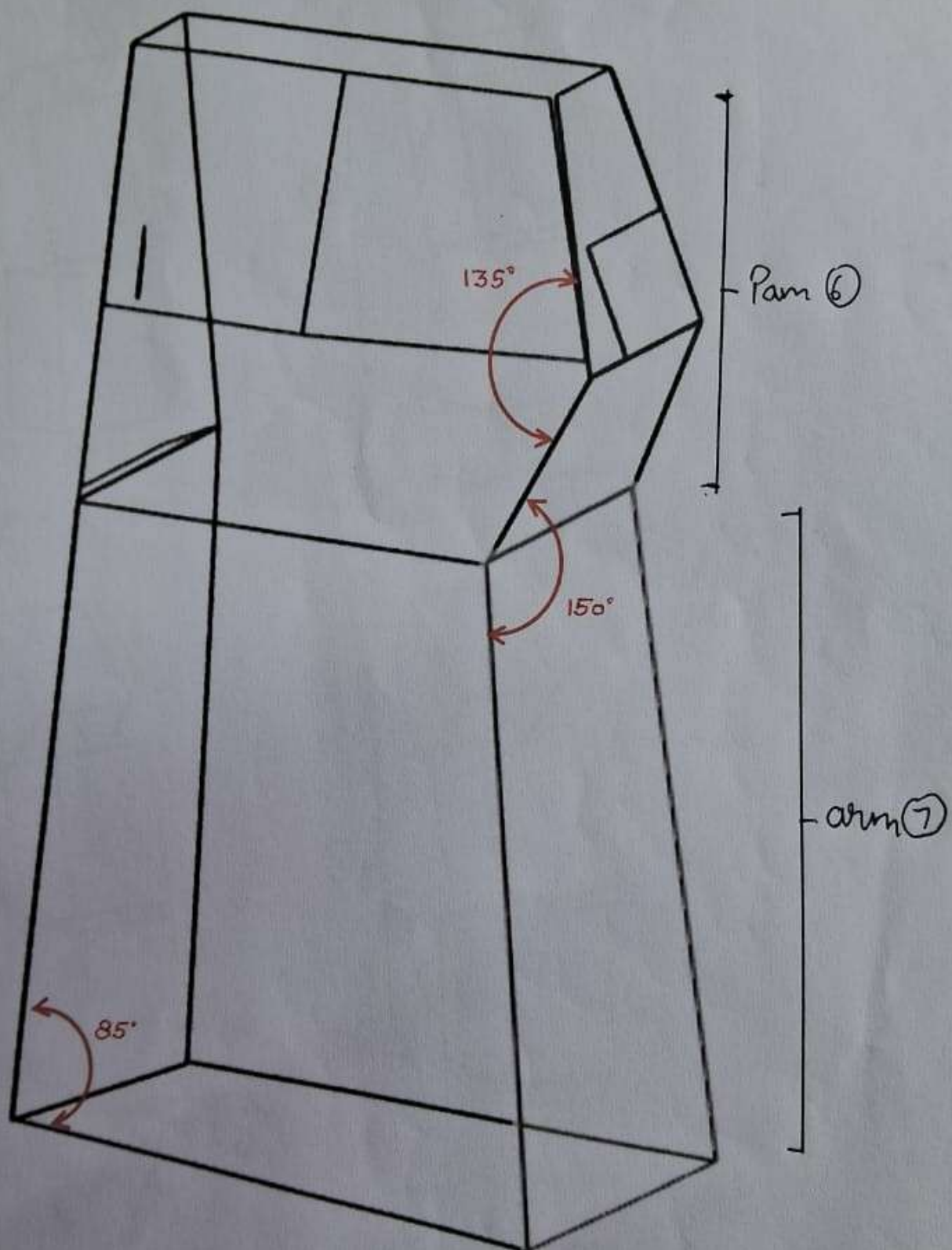
5

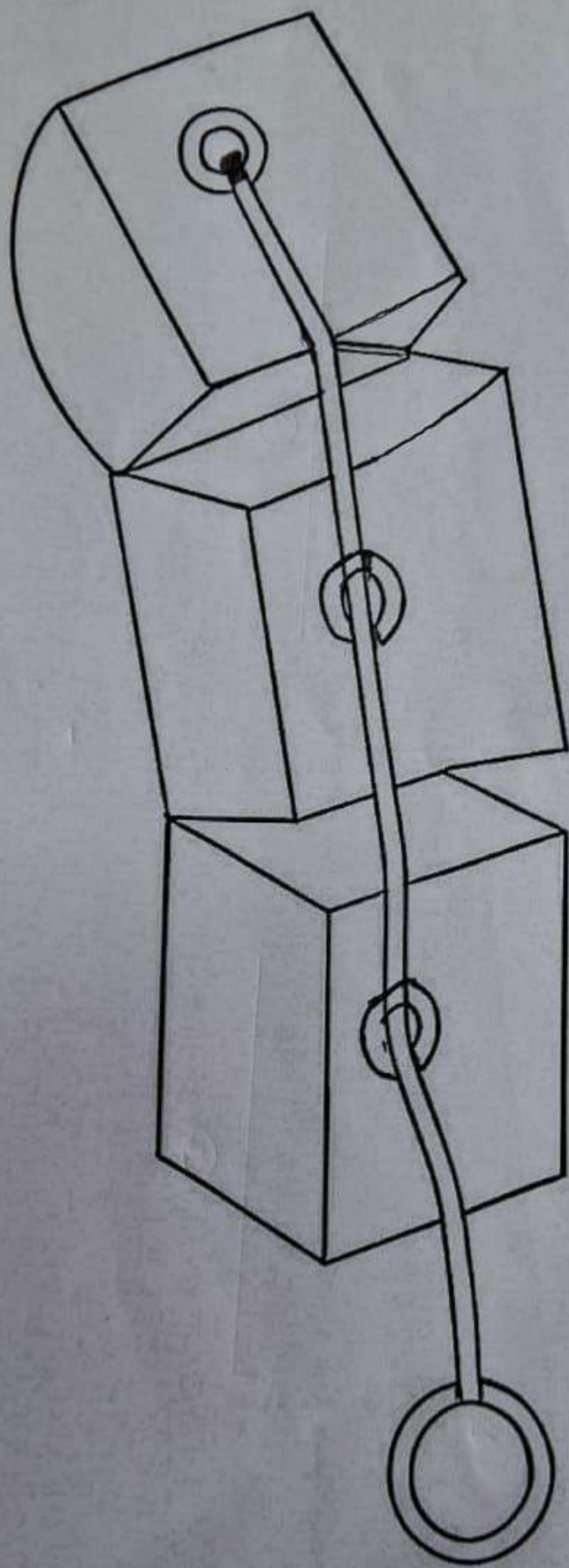
Thumb

⑥ lam

⑦ arm







↓ contraction

